FORMAT FOR PREPARATION OF PROJECT REPORT
FOR
Foundations of Artificial Intelligence

AI-BASED FOOD RECOMMENDATION SYSTEM

A Content-Based Diet and Meal Recommendation Application

A PROJECT REPORT

Submitted by

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BONAFIDE CERTIFICATE

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ABSTRACT

Choosing what to eat is a daily decision that directly affects health, yet many people, especially students and working individuals, rely on random choices, convenience foods or generic diet charts that ignore their dietary preference, fitness goal and calorie needs.

The AI-Based Food Recommendation System is a Streamlit web application that suggests suitable meals based on the user's preferences. The user selects a diet preference, a fitness goal, a preferred cuisine and a meal type, sets a maximum calorie limit per serving and a minimum protein requirement, and lists any ingredients to be excluded.

The system first filters out meals that do not satisfy these conditions. The food descriptions of the remaining meals are converted into numerical vectors using TF-IDF, and cosine similarity measures how closely each meal matches the user's preferences. A goal-based score is then combined with the similarity score to rank the meals, and the best-ranked meals are shown to the user.

The project demonstrates how content-based filtering and simple nutritional rules can be combined to give personalised, easy-to-understand food suggestions through an interactive interface.

CHAPTER 1
INTRODUCTION

Healthy eating is an important part of a good lifestyle, but selecting the right food every day is not easy. A suitable choice depends on factors such as diet type, health or fitness goal, cuisine preference, the type of meal and the nutrients a person needs. Most people do not have access to a dietician and rely on general advice that may not suit them.

The AI-Based Food Recommendation System addresses this problem. It collects a small set of user preferences, removes meals that do not satisfy the selected conditions, and uses a content-based recommendation technique to rank the remaining meals according to how well they match the user's requirements and goal.

1.1 Problem Statement

People often struggle to decide what to eat in a way that suits their diet, taste and fitness goal. Generic diet plans do not consider individual preferences, and searching through large lists of recipes or foods to find meals within a calorie and protein limit is time-consuming. Existing food and recipe applications mainly provide search or popularity-based lists with limited personalisation. Therefore, an AI-based food recommendation system is proposed that accepts the user's diet, goal, cuisine, meal type, calorie limit and excluded ingredients, filters unsuitable meals, and ranks the remaining meals using TF-IDF, cosine similarity and a goal-based score.

1.2 OBJECTIVES

The main objectives of the AI-Based Food Recommendation System are:

- To collect user preferences such as diet preference, fitness goal, preferred cuisine and meal type through an interactive interface.
- To allow the user to set a maximum calorie limit per serving and a minimum protein requirement.
- To allow the user to exclude specific ingredients (for example, ingredients they are allergic to or dislike).
- To filter out meals that do not satisfy the selected conditions.
- To represent food descriptions as numerical vectors using TF-IDF.
- To recommend meals that closely match the user's preferences using cosine similarity.
- To rank the meals by combining the similarity score with a goal-based score.
- To present the recommendations in a simple, interactive Streamlit dashboard.

1.3 SCOPE OF THE PROJECT

The system is intended for individuals, especially students and young adults, who want quick and personalised food suggestions that fit their diet and goal. It covers preference input, constraint-based filtering, content-based recommendation, goal-based ranking and display of results. It is an educational prototype and a decision-support tool for general healthy eating. The nutrition values used are illustrative estimates, and the system is not a substitute for medical or dietetic advice. Users should always check ingredient labels and allergy information before eating.

The system can be extended in the future with larger verified food databases, personal profile-based calorie calculation, meal-plan generation, user feedback and collaborative filtering.

CHAPTER 2
LITERATURE REVIEW

2.1 Recommender Systems

Recommender systems suggest items to users based on their preferences and behaviour. Common approaches include content-based filtering, collaborative filtering and hybrid methods. Content-based methods recommend items similar to those a user prefers, whereas collaborative filtering uses the ratings of similar users. These techniques are widely used in entertainment and e-commerce and form the basis for food recommendation.

2.2 Food Recommendation

Food recommendation differs from the recommendation of movies or products because food is closely linked to health. A good recommendation must consider taste as well as nutrition, dietary restrictions, cultural habits and the time of the meal. Earlier food recommenders mainly depended on ratings or popularity of recipes, which can promote tasty but unhealthy options.

2.3 Content-Based Filtering and TF-IDF

Content-based filtering represents each item by its features and compares it with the user's preferences. For text features such as food descriptions, ingredients and tags, Term Frequency–Inverse Document Frequency (TF-IDF) converts the text into numerical vectors in which words that are distinctive for an item receive higher weights. This approach works even when no rating data from other users is available.

2.4 Cosine Similarity

Cosine similarity measures the angle between two vectors and is commonly used to compare text vectors. The user's preferences (diet, cuisine, meal type and goal) are converted into a query vector, and each food vector is compared with it. Foods with the highest similarity are ranked first.

2.5 Research Gap

Existing systems often handle only one aspect of the problem, such as recipe search, calorie counting or generic diet suggestions. They do not always combine hard constraints (diet, calories, protein, excluded ingredients) with similarity-based and goal-based ranking in a simple interface. The proposed system integrates these steps in a single lightweight application.

CHAPTER 3
PROPOSED SYSTEM

3.1 System Overview

The AI-Based Food Recommendation System gives users meal suggestions that match their diet, goal and nutritional limits. The user enters preferences in the sidebar of the application and clicks the **Recommend food** button. The system filters the food dataset using the selected conditions, converts the food descriptions of the remaining meals into TF-IDF vectors, computes cosine similarity with the user's preferences, combines it with a goal-based score and displays the ranked results.

3.2 Proposed Workflow

User Input → Filtering → Content-Based Recommendation (TF-IDF and Cosine Similarity) → Ranking (Goal Score + Similarity Score) → Display of Recommended Meals

3.3 Architecture Diagram

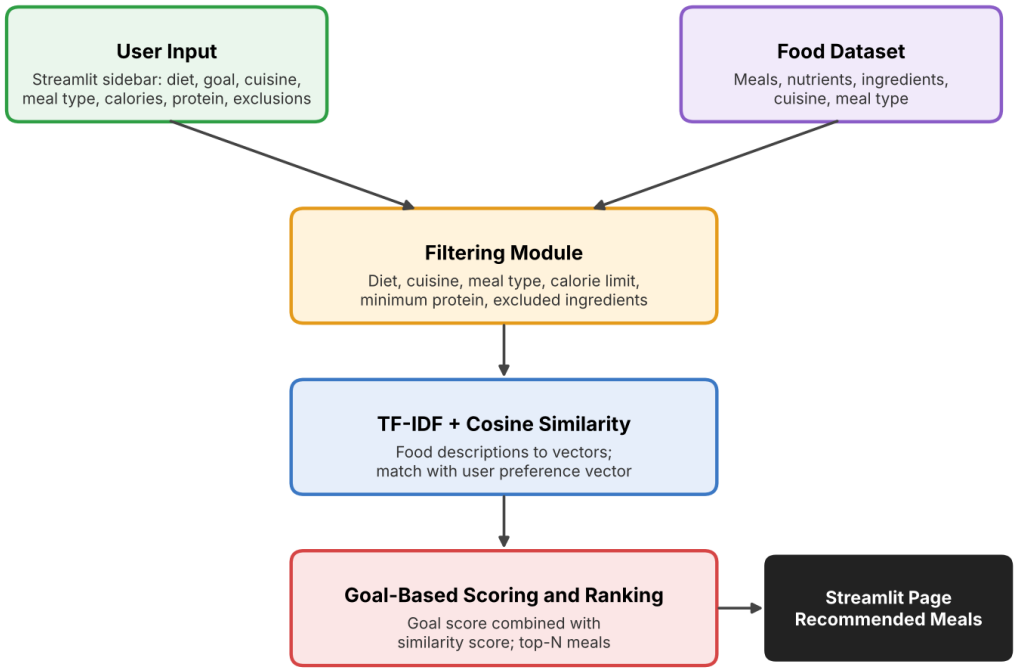


Fig. 3.1 Architecture diagram of AI-Based Food Recommendation System

3.4 Proposed System Workflow

The system follows a sequential workflow, as also described on the home screen of the application:

1 **Input:** The system collects the user's diet preference, fitness goal, preferred cuisine, meal type, calorie limit, minimum protein and excluded ingredients.

2 **Filtering:** Meals that do not satisfy the selected conditions are removed from the dataset.

3 **Content-based recommendation:** TF-IDF converts the food descriptions into numerical vectors, and cosine similarity measures how closely each meal matches the user's preferences.

4 **Ranking:** A goal-based score is combined with the similarity score to rank the suitable meals, and the top meals are displayed.

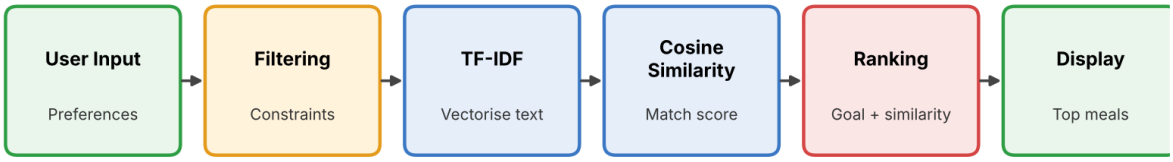


Fig. 3.2 Workflow of AI-Based Food Recommendation System

CHAPTER 4
MODULES DESCRIPTION

4.1 MODULE 1 – USER INPUT AND CONSTRAINT FILTERING

This module collects the user's preferences through the sidebar titled "Your preferences" and removes meals that do not satisfy them.

Inputs collected:

- **Diet preference:** Any, or a specific diet such as vegetarian or non-vegetarian.
- **Your goal:** The fitness goal, for example Balanced diet.
- **Preferred cuisine:** Any, or a specific cuisine.
- **Meal type:** Any, or a specific meal such as breakfast, lunch, dinner or snack.
- **Maximum calories per serving:** A slider that sets the upper calorie limit (700 kcal in the default view).
- **Minimum protein (g):** A slider that sets the minimum protein per serving (10 g in the default view).
- **Excluded ingredients:** Ingredients that must not appear in the recommended meals.

Main Functions:

- Collection of user preferences through dropdowns and sliders
- Filtering by diet preference, cuisine and meal type
- Filtering by maximum calories and minimum protein
- Removal of meals containing excluded ingredients

4.2 MODULE 2 – CONTENT-BASED RECOMMENDATION AND RANKING

This module recommends and ranks meals from the filtered dataset. The text description of every food item (such as its ingredients, cuisine, meal type and tags) is converted into a TF-IDF vector. The user's preferences are converted into a query vector in the same way, and the cosine similarity between the query vector and every food vector is calculated.

A goal-based score is then calculated for each meal according to the selected goal. For example, a goal that favours higher protein gives a higher score to protein-rich meals, while a balanced-diet goal favours meals with a balanced nutrient profile. The final ranking score combines the two scores:

$$Final\ Score = (w \times Similarity\ Score) + ((1 - w) \times Goal\ Score)$$

Meals are sorted by the final score and the top-ranked meals are displayed to the user.

Main Functions:

- Food description vectorisation using TF-IDF
- Cosine-similarity-based matching with user preferences
- Goal-based scoring of meals
- Combined ranking of meals and selection of top-N results

4.3 MODULE 3 – USER INTERFACE

The interface is built with Streamlit. The sidebar contains the preference inputs and the **Recommend food** button. The main page shows the application title, a short description, an instruction message, an explanation of how the AI system works and the recommendation results. A disclaimer states that the project is an educational prototype, that the nutrition values are illustrative estimates and not medical or dietetic advice, and that users should check ingredient labels and allergy information before eating.

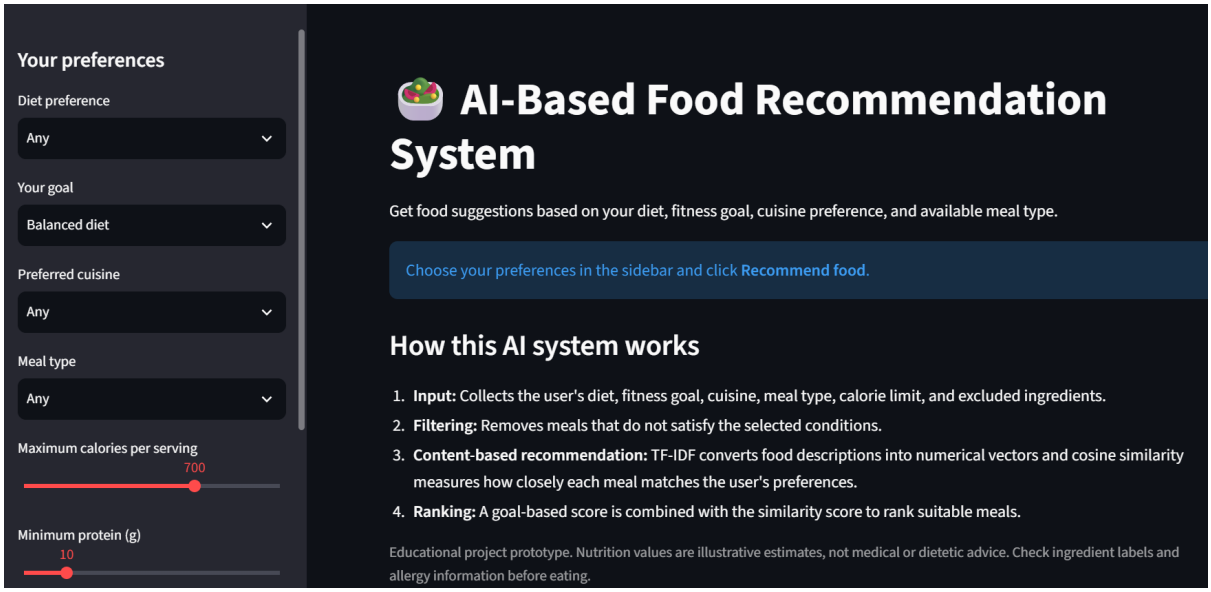


Fig. 4.1 Home Screen and Preference Sidebar of the Application

CHAPTER 5
IMPLEMENTATION AND RESULTS

5.1 EXPERIMENTAL SETUP

The AI-Based Food Recommendation System was implemented as a Python-based Streamlit application. It was tested using a sample food dataset containing nutritional values, ingredients, cuisine and meal-type information. Each module was tested individually to verify its functionality and output.

5.1.1 Software and Technologies

Component	Technology
Programming Language	Python
Data Processing	Pandas, NumPy
Text Features	TF-IDF (Scikit-learn)
Recommendation Technique	Content-based filtering, Cosine similarity
Machine Learning Library	Scikit-learn
User Interface	Streamlit
Development Environment	VS Code

5.1.2 Functional Testing

The major functions of the system were tested using sample inputs.

Test Case	Expected Output
Preference Input	Diet, goal, cuisine and meal type selected from the sidebar
Calorie Limit Filter	Only meals within the maximum calories per serving retained
Protein Filter	Only meals meeting the minimum protein retained
Diet / Cuisine / Meal-Type Filter	Only matching meals retained
Excluded Ingredient Filter	Meals containing excluded ingredients removed
Content-Based Recommendation	Similarity score calculated for each remaining meal
Ranking	Meals ordered by combined goal and similarity score
Recommend Food Button	Ranked list of recommended meals displayed

5.2 RESULTS

The implemented system provides an interactive interface for personalised food recommendation. The results include the filtered meals and their ranked recommendations according to the user's preferences.

5.2.1 Home Screen and Preference Input

When the application starts, the user sees the sidebar with the preference inputs and a message asking them to choose preferences and click **Recommend food**. The main page also explains how the AI system works (see Fig. 4.1).

5.2.2 Filtering Results

After the user clicks **Recommend food**, meals that do not satisfy the diet, cuisine, meal-type, calorie, protein and excluded-ingredient conditions are removed.

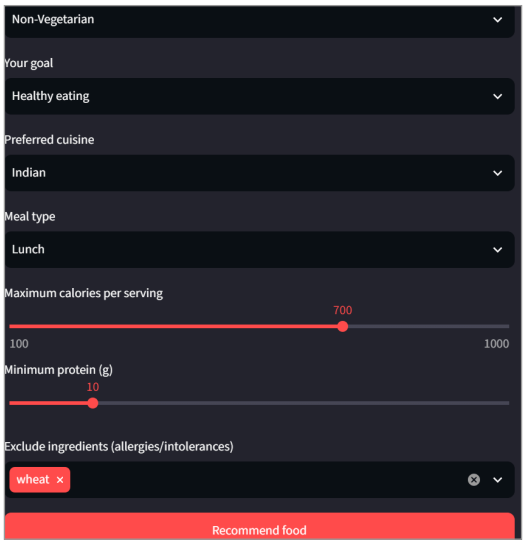


Fig. 5.1 Filtered Results for a Sample Set of Preferences

5.2.3 Recommendation Results

The ranked list of recommended meals is displayed with their nutritional details and match scores.

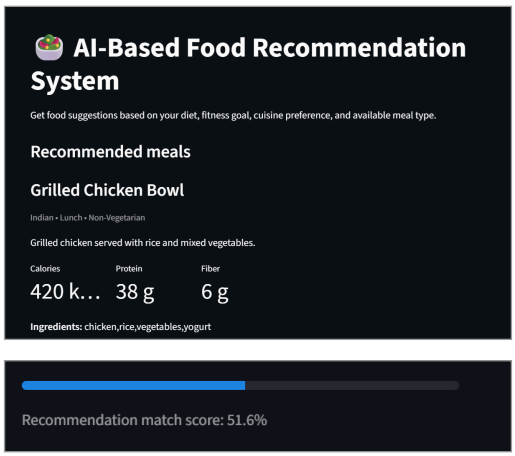


Fig. 5.2 Recommended Meals for a Sample Set of Preferences

CHAPTER 6
CONCLUSION AND FUTURE WORK

6.1 Conclusion

The AI-Based Food Recommendation System is an interactive application that helps users choose suitable meals according to their diet, goal, cuisine, meal type, calorie limit and protein requirement. It combines constraint filtering, TF-IDF text vectorisation, cosine similarity and goal-based scoring to rank meals and present the most suitable ones.

The project shows how content-based filtering, a technique from Artificial Intelligence, can be applied to everyday food decisions even without rating data from other users. The Streamlit interface makes the system simple to use, and the modular design makes it easy to extend.

6.2 Future Work

The future development of the system can include:

- Integration of larger and regional food databases with verified nutrition data.
- Personal profile input (age, gender, height, weight, activity level) with BMI and calorie-requirement calculation.
- Allergy-aware filtering with a wider list of allergens.
- Daily meal-plan generation within a calorie target.
- Nutrition analytics with charts for calories, protein, carbohydrates, fat and fibre.
- User ratings and feedback to improve recommendations.
- Collaborative and hybrid filtering using ratings from many users.
- Explainable recommendations showing why a food was suggested.
- Support for medical conditions with expert validation.
- Multilingual and voice-based assistance.

CHAPTER 7
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